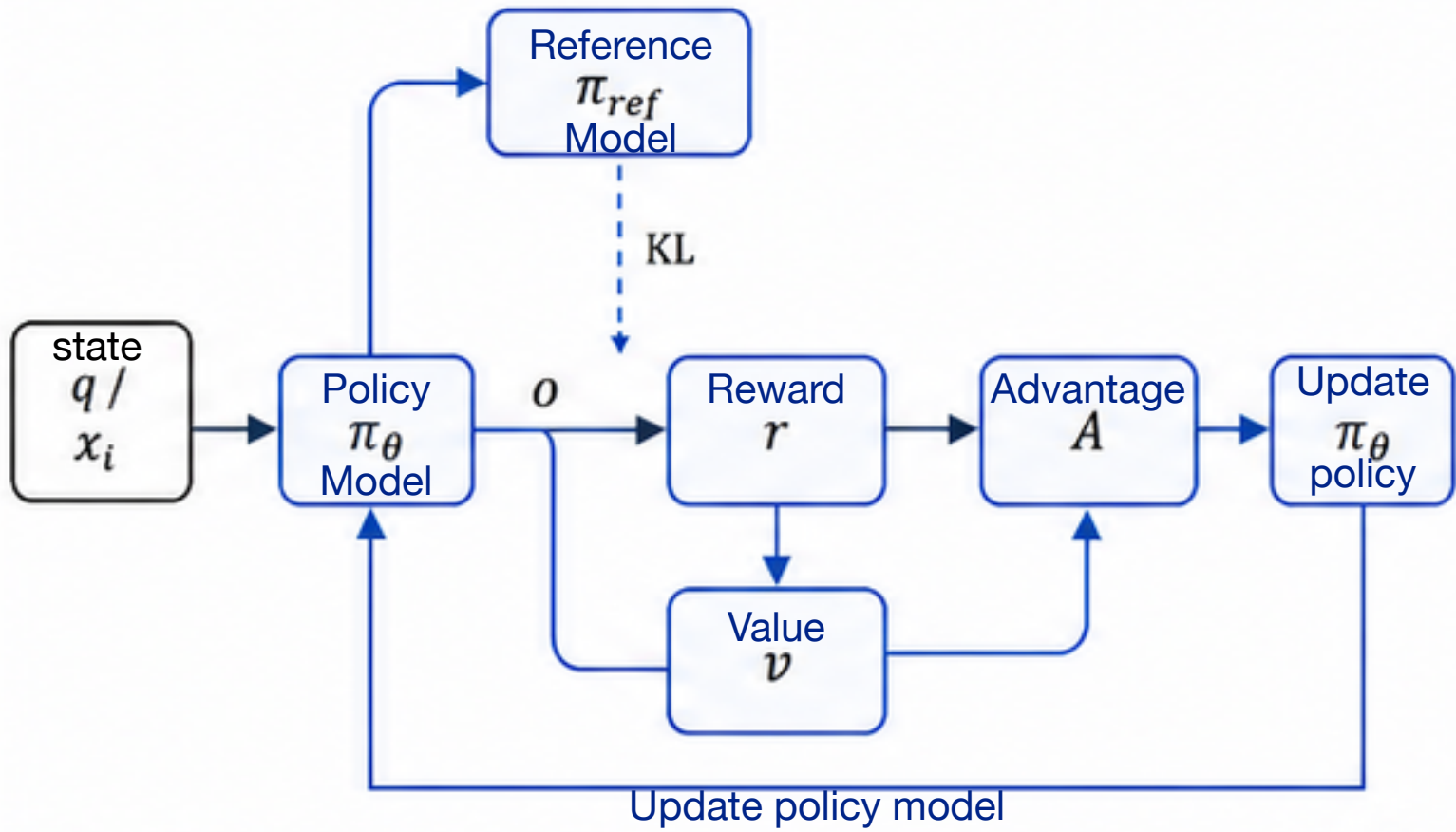
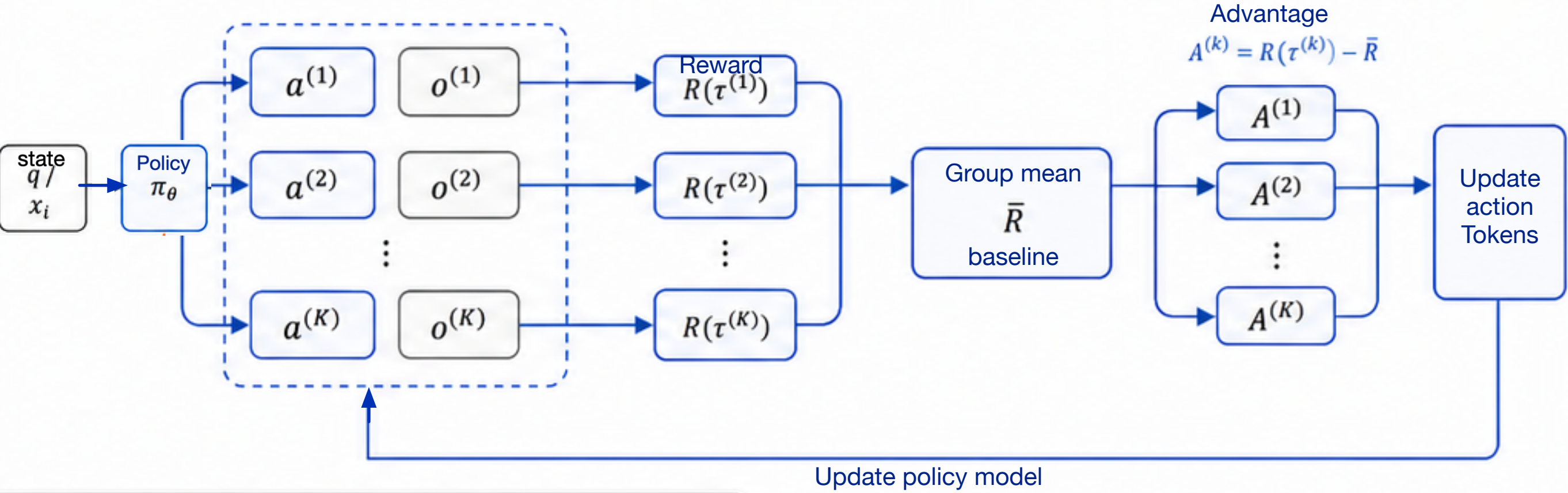


PPO: value-based policy optimization

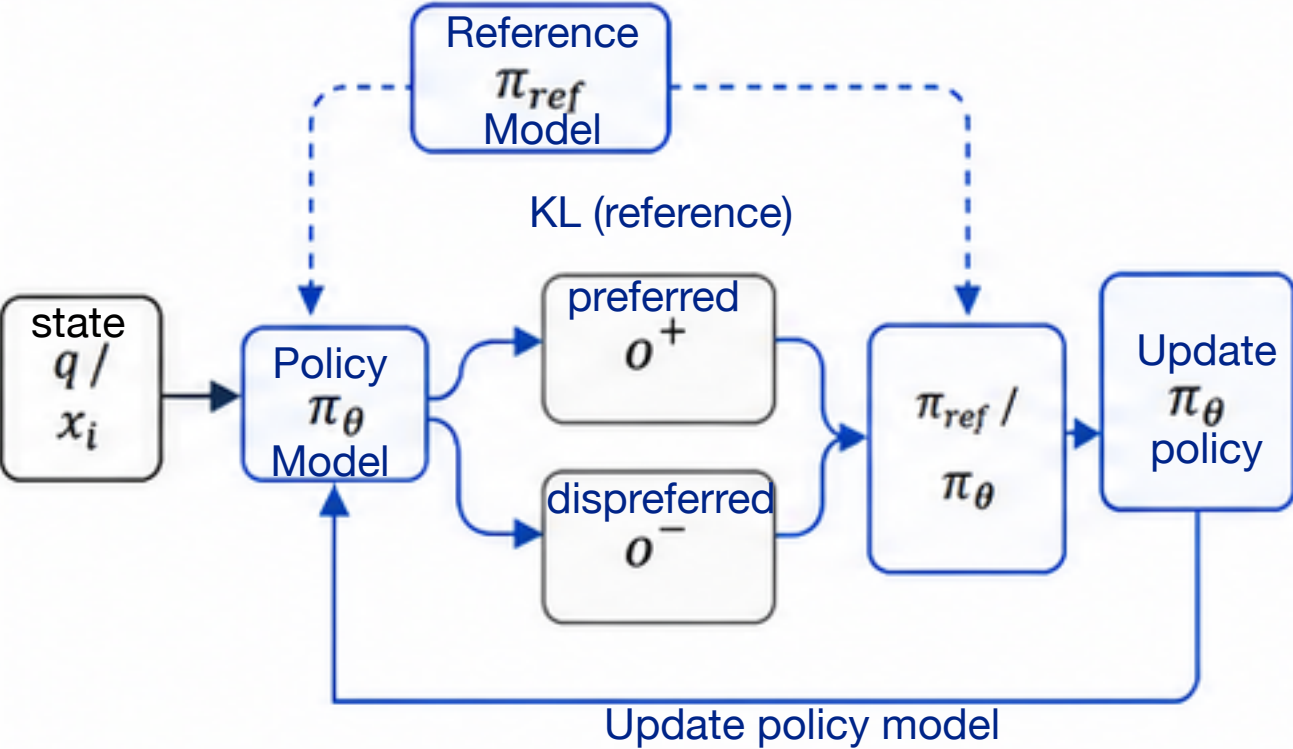


GRPO: action-only group optimization



Same GRPO optimizer; changed trajectory parameterization

DPO: preference-pair optimization



ReGRPO: gated reflection-correction trajectories

